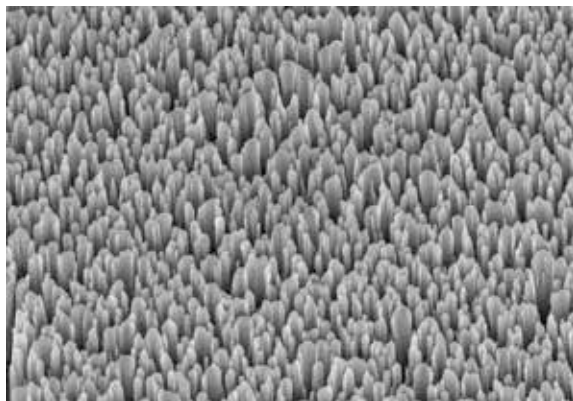


Credit: Harvard



One dimensional germanium nanowires grown on a silicon substrate

whereas a radius of 3 m has a ratio of $3/3 = 1$.

Nano-coatings: Reducing a material to a nanoscale in only one dimension results in a nano-coating, nanoribbon or a nanosheet, which is normal sized in all dimensions but thickness. This is typically used to coat surfaces in order to protect them from scratches, or to make them water or dirt repellent, or to add on anti-glare (for spectacles) or change opacity, or to strengthen them, such as with nanocomposite polymers.

Nanowires: Also known as nanofibres (depending on rigidity), have two dimensions in the nanoscale. Carbon fibre is an example, and can be woven into things in order to add on stability and durability. In nature, plants use cellulose, which is an organically produced nanofibre, to give their cell walls the rigidity they need. In nanotech, nanowires are produced using a variety of methods that are either top down or bottom up manufacturing (explained later). They are usually made with etching, or chipping away at a larger wire, or using the now famous VLS (vapour-liquid-solid) method from the 1960s that produced silicon whiskers using droplets of gold. Nanowires are critical to all of us geeks, because they are what make the miniaturisation of electronics possible.

Nanotubes: Like a pipe, but a billion times smaller, a nanotube is a feat of modern engineering. It's actually made up of atoms bonding together to form a tube so small, that it's about as thin as a nanowire, but hollow! Nanotubes have very

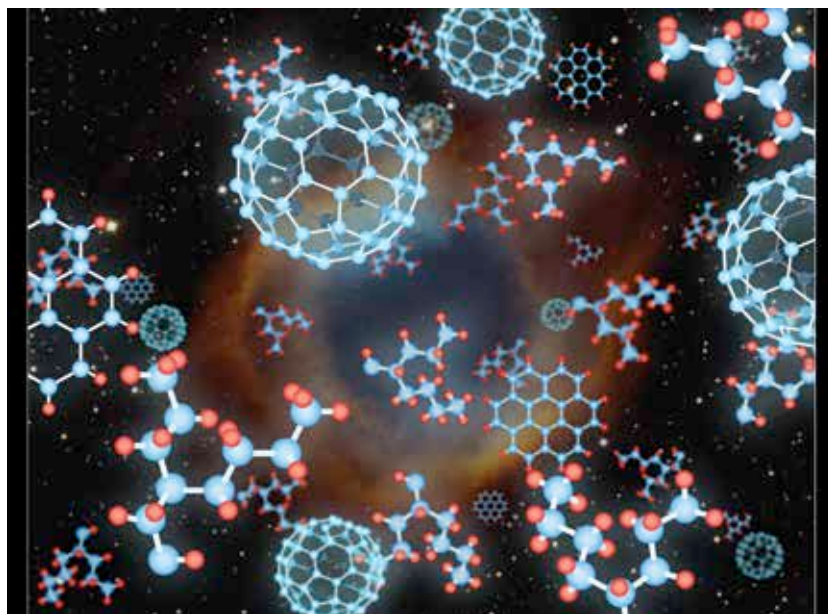
interesting properties of conductivity depending on how they're made and with which materials. They're able to conduct or insulate (depending on material), along their surfaces. This makes them perfect for some super advanced use cases, such as nanoscale targeted drug delivery

methods, chemical sensors that are sensitive to a few parts per billion (or more!), very, very precise thermal conductivity and monitoring.

Nanoparticles: While one usually thinks of small amounts of a substance as atoms or molecules, and thus a nanoparticle would seem like just a tiny amount (few atoms or molecules of a substance), they're actually a little more than that. For our purposes, we're obviously interested in human-engineered nanoparticles, and not the nano-sized particles that nature produces. While mostly used to enhance chemical properties of materials or to act as catalysts, they are also used extensively to create what are called nanocomposite polymers – think of these as polymers that have layers and structures that are precise at even a nanoscale

with their composite mixtures. Composites are part of everyday life for us, when it comes to plastics, or silica-infused tyres that last longer, food packaging that repels moisture, creating materials that are not flammable, and more.

Quantum dots: Essentially a subset of nanoparticles, quantum dots are nanoscale semiconductors that are very popular in modern display technology. The way they behave is because of their tiny, quantum-scale size, which allows quantum mechanics to rule over classical mechanics. Also known as “artificial atoms”, quantum dots are a very interesting material science development, because they can be made to emit light of different wavelengths (colours) depending on the size of the dots, and their composition or shape. While the most obvious use for the optical properties of quantum dots is in the display industry (if you have heard the term before, it's most likely to be because of ads for TVs), there are plenty of other areas where they are being used as well. The second most obvious use case is lighting, but we're not talking about LEDs and lighting your home, we're talking about lighting materials and liquids and doing so at nano scales, so as to get extremely even lighting of slices that are nano-thin. Biology labs use dyes to mark growth of cultures



Credit: Pete Marenfeld (NOAO)

Examples of some buckyball structures. These fullerenes are strong and hard like diamonds.